Histopathology of Infection of *Crassostrea virginica* (Gmelin) by *Dermocystidium marinum* Mackin, Owen, and Collier

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ALTHOUGH MASS MORTALITIES of oysters suspected or known to be caused by disease have been reported by various authors (Needler, 1931; Orton, 1924; Korringa, 1947; Roughley, 1926), at widely separated points in the world, few studies have been made of the pathology of infection by a known organism. Roughley (1926) described and figured abscesses in diseased tissues of oysters without determining whether or not an invading organism was present. Prytherch (1940) described degenerative effects of adductor muscle fibers resulting from heavy infections by *Nematopsis ostrearum*, a gregarine parasite. Orton (1924) recorded a high percentage of pale-colored livers, and discussed myolysis in diseased oysters but without coming to any conclusion as to the cause of the muscular degenera-

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All figures are photomicrographs

Fig. 1. *D. marinum* in adductor muscle of oyster, to show nature of parasite. This section was selected because of the few leucocytic cells present and shows the spherical nature of the parasitic cells, the eccentric vacuole and inclusion body contained therein. Lysed cavities are evident. Delafield's haematoxylin and eosin. Original magnification about 950 diameters.

Fig. 2. Inflammation of the intestinal epithelium and connective tissues underlying the basal membrane due to concentrations of leucocytic cells. Only moderate infection of the epithelium and few parasites have invaded the connective tissue. Delafield's haematoxylin and eosin. Original magnification about 400 diameters.

Fig. 3. Normal intestinal epithelium. This oyster was parasitized locally (see Fig. 4) and some tissue response and congestion is to be seen in the connective tissues. Harris' haematoxylin and eosin. Original magnification about 100 diameters.

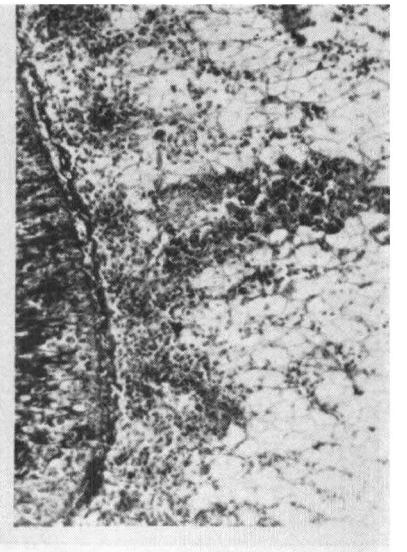
Fig. 4. Abscessed intestinal epithelium of the same oyster as that shown in Fig. 3, the section through a different loop. Note thickening of the epithelium, the numerous characteristic round cavities containing the parasitic cells, separation of the basal membrane from the underlying connective tissue, destruction of the ciliated border and leucocytic infiltration. Harris' haematoxylin and eosin. Original magnification about 100 diameters. Compare with Fig. 3.

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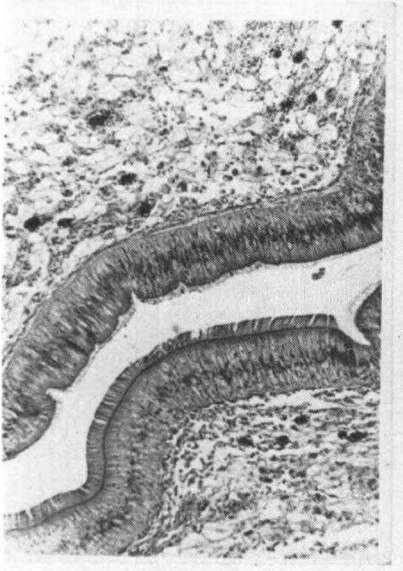
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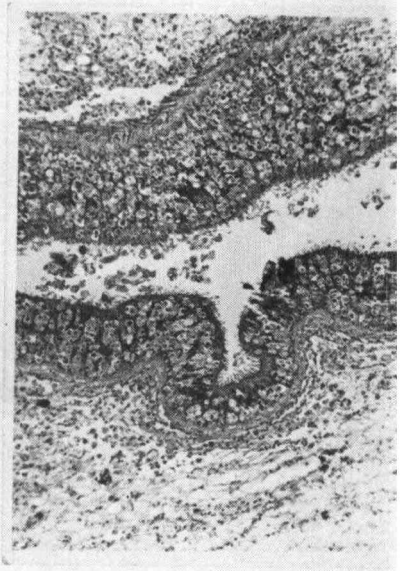
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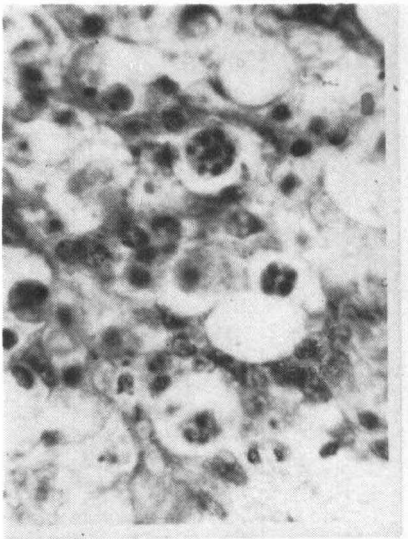
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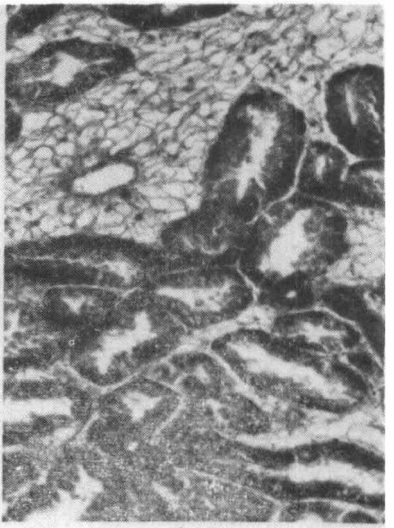
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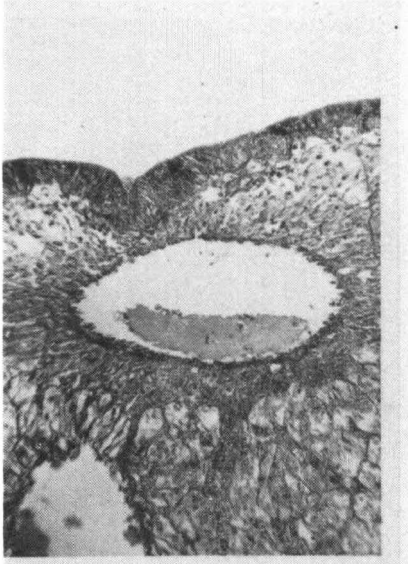
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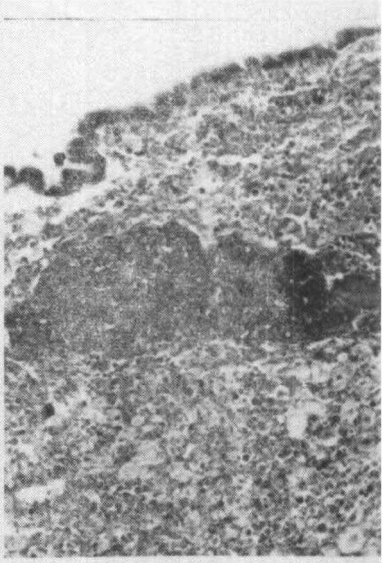
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tion. He also noted pathological conditions in stomach epithelium and in connective and reproductive tissues, as well as cyst formation. Although Orton failed to find a cause for the pathological conditions described, he figured what are undoubtedly parasitic cells closely resembling certain Sporozoans. His figures also show extreme leucocytosis in tissues underlying the stomach epithelium and in blood spaces, but he did not discuss the pathological processes related to such concentrations. In one figure Orton shows "blood cells" which are smaller in average size than the nuclei of adjacent cells, which prompts the suspicion that he actually observed concentrations of parasites. Korringa (1948) described hypertrophy and consequent abnormal secretion of cells of the mantle epithelium induced by a shell-boring fungus. Giard (1894) in discussing the "maladie du pied" called attention to the degeneration of the shell attachments of the adductor muscle fibres, and attributed the general atrophy to attacks of a Schizomycete, *Myotomus ostrearum*, a conclusion somewhat doubtful in view of the nature of the studies made. Needler and Logie (1947) described abscesses, tissue breakdown, and general emaciation resulting from a disease of unknown etiology in oysters of Prince Edward Island, Canada. In general, details have been lacking in all of these studies and in most cases either no associated parasite was indicated or the etiology was in doubt. In no case is there evidence that systematic comparison of pathologic conditions with normal tissues was a part of the studies.

Fig. 5. A small section of diseased epithelium from the abscessed area shown in Fig. 4 more highly magnified. Note the lysed cavities and reproductive stages of the parasite. Original magnification about 950 diameters, greatly enlarged.

Fig. 6. Section through a portion of the digestive gland of a normal oyster with no infection, showing tubule and supporting connective tissue with a small blood vessel. Heidenhain's haematoxylin and eosin. Original magnification about 400 diameters.

Fig. 7. Section through the margin of the mantle of a normal, uninfected oyster to show one of the large blood vessels with radiating fibrous connective tissue, epithelium, and Leidig-cell basal connective tissue. Heidenhain's haematoxylin and eosin. Original magnification about 220 diameters.

Fig. 8. A portion of necrotic mantle to show amorphous basophilic mass resulting from what appears to be liquefaction of tissue resulting from combined auto-digestion of disintegrating cells and lytic action of larger numbers of parasites. Delafield's haematoxylin and eosin. Original magnification about 220 diameters.

This paper presents data on the pathologic changes in various tissues of *Crassostrea virginica* associated with infection by *Dermocystidium marinum* Mackin, Owen, and Collier (1950) a recently discovered Protistan. It is characteristic of this parasite that massive infections are built up in the host. Large numbers of oysters in Barataria Bay, Louisiana, and adjacent areas are infected during high temperature months, providing an excellent opportunity, because of abundance of material, to make a detailed study of histopathological changes related to various stages of infection.

Researches on the diseases of oysters, as well as on many other phases of oyster biology, have been supported by The Texas Company, Humble Oil and Refining Company, The California Company, Tidewater Associated Oil Company, Phillips Petroleum Company, Shell Oil Company, and Gulf Refining Company. The author wishes to make grateful acknowledgement not only for past and continuing financial support, but also for helpful collaboration in every possible way. Due also to my colleagues of the Texas A.&M. Research Foundation is acknowledgement of active collaboration. Dr. Sewell H. Hopkins has carried the burden of library search and direction and counsel. Fred Cauthron and Joe Prokop have made all photomicrographs, and slide techniques and related laboratory work has been largely the work of Dan Wray and J. L. Boswell. Many others have contributed materially. Interpretations of the data are the responsibility of the author.

MATERIALS AND METHODS. Sections of more than 1000 oysters have been made and studied. Many of these are not usable for histologic study because of degeneration of tissue at death of the host. Oysters used came from several sources. These were as follows: (1) from 17 oyster experimental field stations in Barataria Bay, Louisiana, and adjacent waters of the same region, (2) from laboratory experimental tanks at the Grand Isle Laboratory of the Texas A.&M. Research Foundation, (3) from various planted oyster beds in Barataria Bay and vicinity, (4) from several points on the Atlantic coast of the United States and points on the Gulf Coast outside of Louisiana. These include oysters from Delaware Bay (New Jersey); Rappahannock and James rivers, Virginia; Charleston, South Carolina; Pensacola, Florida; Gulfport, Mississippi and Aransas Bay, Texas. In addition, sections of *Ostrea edulis* (France and Holland), *Ostrea equestris*

(Texas Gulf Coast), and *Crassostrea commercialis* (Australia) have been studied. Outside of Louisiana Dermocystidium has been found only in the Virginia rivers, South Carolina, Florida, and Mississippi.

Fixation of oysters has been by (1) Zenker's fluid, (2) Bouin's fixative, (3) 10% formaldehyde, (4) Schaudinn's fluid, and (5) Helly's fluid. Only Zenker's fluid gave consistently satisfactory results. Oysters in nearly all cases were fixed *en toto;* those cut into smaller pieces before fixation showed little advantage. Transverse blocks were cut from the fixed oysters and sectioned and a variable number of slides were made from each block; the sections were oriented to give a transverse section of the visceral region in the vicinity of the base of the palps. Some of these included palp tissue and some gill tissue, some both types, in addition to others which would always be included in such a section. Many oysters were sectioned through the ventral part of the adductor muscle and tissues ventral to the muscle, which included the lower loops of the intestine and style sac, renal epithelium, and posterior margin of the gonads. A special study was made of the heart and visceral ganglion. Additionally many sections were made of other organs or regions for special purposes.

Both living and dead oysters were fixed and sectioned. The live ones varied widely in concentration of infective elements from negative, of which there were many, through light, moderate, and heavy infections. Many of these "live" oysters were actually gapers (i.e., too weak to effect closure of the shell by the adductor muscle) but with ciliary movement, mantle sensitivity, or weak heart beat to attest to the living condition of the tissues. These still-living gapers provided the best material for study of the effect of massive infections. For experimental purposes such oysters are classed as dead, for they never recover if heavily infected with Dermocystidium. Oysters actually dead when recovered are of little value for histological study, since it is impossible to make interpretations from dead tissues. Many such oysters were sectioned, and most were in good enough condition to allow determination of the degree of infection and were valuable in that connection. Some few were so far disintegrated by the action of saprophytic bacteria that they were useless for any purpose.

Several hundred oysters with moderate to heavy infections were stained with a Gram stain to determine the incidence of bacterial invasions. These latter were rare and were associated with degenerative

changes taking place after death. A few oysters showed actual parasitic invasions with bacteria as complications of infection with *D. marinum*. Such complications were easily separable in effect from the primary infections. A few cases of primary bacterial parasitization of live oysters without *D. marinum* infection were recorded. Additionally several other diseases of oysters are present in Barataria Bay and appeared in sections in various percentages of the specimens.

NATURE OF THE PARASITE. A preliminary description of *D. marinum* has appeared in Science (Mackin, Owen, and Collier, 1950). It need only be noted here that the parasite is believed to be a fungus and the only known stages are small spherical vacuolated cells (Fig. 1), which reproduce prolifically in the oyster tissues by a schizogony-like process closely resembling that of certain Sporozoans. The cells of Dermocystidium are morphologically somewhat like those of the human blastomycosis-producing organisms such as Blastomyces and Cryptococcus. Infective stages are not known, nor has production of anything resembling a mycelium been observed. So far as now known *Crassostrea virginica* is the only host.

HISTOLOGICAL CHANGES ASSOCIATED WITH *Dermocystidium marinum* INFECTION IN *Crassostrea virginica*. For purposes of discussion three stages of infection with *D. marinum* may be recognized. Light infections are defined as those in which the distribution of parasitic cells is such that search is necessary to confirm infection. A moderate infection is defined as one in which local foci of infection have developed to the point where a number of parasitic cells are concentrated in localized areas of tissue, producing obvious histologic changes, but in which a large part of the host tissue has not been infiltrated. A heavy infection is one where the major part of all tissues excepting certain epithelia are more or less heavily infiltrated and at least some areas contain massive infections.

Several basic characteristics of *D. marinum* infections should be noted at this point. First, any tissue is susceptible to invasion although even in heavy infections the external epithelia and peripheral nerves are not usually penetrated. Second, infection during high temperature months and in high salinity waters proceeds to the point where it can only be described as massive. In development of massive infection as well as in some other respects Dermocystidium disease resembles human blastomycosis (Gilchrist's disease). For an account of this

latter see Conant, *et al.*, (1944). Third, the damage done to host tissues results largely from lytic action of the parasitic cells, coupled with embolism in late stages. There is apparently no marked toxic action judging from the sometimes extreme concentrations of parasites built up before death of the host.

Light infections of *D. marinum* are usually to be found only in the digestive epithelium and may be distributed from the stomach to the rectal region. In some slides, occasional cells may be found in the mantle connective tissue when there are none in the digestive epithelium, but it is considered likely that in these cases sections of the digestive tract in body regions other than the standard anterior visceral location would have shown a primary infection. However, this is by no means a certainty in all cases and it is distinctly possible that infection may take place through any epithelial surface.

The first reaction to infection is a marked leucocytosis and the usual migration of phagocytic cells to the site of parasitic lodgement, resulting in inflammation (Fig. 2). Leucocytes invading the digestive epithelium often are in such numbers that displacement of columnar epithelium may be extensive as shown in Figure 2. Masses of leucocytic cells accumulate in the connective tissue under the basement membrane even when few parasites have migrated from the digestive epithelium.

After establishing a primary focus in the digestive epithelium, an extensive abscess may be produced by development of large numbers of parasites. Normal intestinal epithelium is composed of tall columnar cells, closely packed and with nuclei distributed near the middle. The ciliated border is sharply defined (Fig. 3). Extensive infiltration of the epithelium results in (1) infiltration by leucocytes (2) destruction of epithelium and leucocytes alike by lytic action of the parasites and (3) destruction of ciliated border (Fig. 4). Separation of the basal membrane from the underlying connective tissue often takes place and the resulting cavity fills with leucocytic and also parasitic cells if the basal membrane has been penetrated by the latter (Fig. 4). Figure 5 is a greatly enlarged portion of Figure 4 showing the development of parasites in lysed cavities characteristic of *Dermocystidium marinum* infection.

Such heavy concentrations as shown in Figure 4, ultimately result in liberation of the parasites into the lumen of the digestive system. Large areas of the epithelium are sloughed off from the basement

membrane, and the necrotic tissue and parasites fill the lumen. They also gain access to the connective tissues through lysis of the basement membrane. Once in the connective tissue mesenchyme, distribution to all parts of the body takes place through the medium of the blood sinuses and large veins and arteries. In advanced stages of disease numerous parasites and leucocytes may completely obliterate the sinuses and obstruct all but the largest blood vessels.

The connective tissue, especially in the digestive gland region, mantle, and gills, is peculiarly susceptible to attack. Connective tissues in the oyster consist of large Leidig cells, which are derived from modified leucocytic cells, and localized areas of fibrous tissues, which are especially marked about the walls of blood vessels. In *Crassostrea virginica* the yellowish "pigment" cells are sparingly distributed throughout the connective tissues but are concentrated in the auricle of the heart and to a lesser extent about the basal membrane of the digestive system throughout its length. Figure 6 shows connective tissue supporting tubules of the digestive gland in a normal oyster. A small amount of fibrous tissue is radiating from the small blood vessel. Figure 7 shows normal connective tissue composed of radiating fibers and Leidig cells around a large blood vessel in the mantle.

Infiltration of connective tissues results in either (1) generalized evenly distributed infective elements ultimately resulting in conversion of the entire visceral mass into one large abscess (Figs. 8 and 9), or (2) more localized infection producing small abscesses (Figs. 10, 11, and 12). This latter condition is often encountered in the mantle. Sections here often show numerous pinhead-size abscesses filled with tissue debris, parasites, and leucocytic cells. These tend to lie close beneath the mantle epithelium and to destroy tissues of the basement membrane resulting in atrophy, necrosis, and ultimate sloughing of the epithelium (Fig. 11), a process which converts abscess to ulceration.

Several specific pathologic responses to infection of the connective tissues should be noted. First is the usual development of marked fibrosis especially well shown in Figures 11 and 12. In late stages much of this fibrous material may be lysed and the fibroblasts destroyed. Second is the production of a caseous-like deposit resulting from liquefaction of the host tissues in the abscess (Figures 8, 10, and 12). The staining reaction of this amorphous material is often distinctly baso-

philic. Third is the tendency of the so-called pigment cells to increase in numbers. These cells are probably of leucocytic origin and contain deposits of yellowish granules. During infection with *D. marinum* their number is greatly augmented and they appear in tissues and areas where they are normally rare (Fig. 9). The granules stain black with Heidenhain's haematoxylin and dark green with Giemsa. Augmentation of pigment cells has become a valuable pathologic index. The nature of the granules is unknown but it is suggested that it may be some type of metallic deposit of the general nature of haemosiderin.

Phagocytic activity of leucocytic cells results in ingestion of large numbers of parasites which grow and reproduce intracellularly as well as in lumena of cavities. It is common to find parasites in well over 50% of the leucocytic cells which invade infected tissues. Ultimate destruction of all parasitized leucocytes is probable and it is certain that few escape. *D. marinum* cells develop vigorously in the cytoplasm, pushing the nucleus to the periphery, and the cell is represented finally by only a thin shell which ruptures to liberate the parasites. Much of the cellular debris seen in tissue abscesses results from breakdown of leucocytes. In heavy infections diagnosis may often be accomplished by examination of fluid from the pericardial cavity, which may contain enough cells to produce a decided turbidity. Parasites are clearly seen in fresh unstained mounts of such fluid as dark bottle-green rounded inclusions in the moribund leucocytes. Much tissue debris is also contained in such material (Fig. 13).

Concentrations of parasites may often also be found in the fibrous wall of blood vessels, especially of the mantle. Increased deposit of fibrous tissue and invasion by leucocytic cells is the first response to the presence of parasites, and an abnormal thickened appearance of the wall of the vessel results (Fig. 12). Ultimately this tissue is lysed to the extent that the vessel as such no longer exists as a containing vessel and it is difficult to distinguish a cross sectional view from an abscess. Although the heart itself is often affected, complete breakdown of the walls is unusual, and in general the trabeculae are apt to be more lightly infected than are the fibrous walls of peripheral vessels. Circulatory failure is probably one of the most common immediate causes of death of the host.

Transportation of infective cells from primary foci to the adductor muscle is probably via the posterior aorta. Massive infection of part or

all of the adductor muscle is a common occurrence. Longitudinal sections through the muscle give the best picture of the gross pathological results of such infections. Figure 14 is a section of normal bundles of fibers with intervening blood sinuses. Figure 15 shows a section of heavily parasitized muscle with masses of infiltrated leucocytes. Apparently the primary action is destruction of the fiber sheaths and terminal attachments, and individual fibers lie in haphazard and twisted masses. Individual released fibers become the myolytic spindles of Orton (1924) and are often carried by the blood stream to remote parts of the viscera. A muscle such as shown in Figure 15 is soft and mucoid to the touch, and heavily infected parts retain little or no attachment to the shell. Extensive involvement of the adductor muscle results in gaping. Muscles of the mantle are also subject to lysis.

Marked effect of infection on tubules of the digestive gland always results from infection of the supporting connecting tissues. Epithelia of the tubules themselves are not usually attacked, but destruction of the connective tissues and blood channels causes atrophy of the tubule

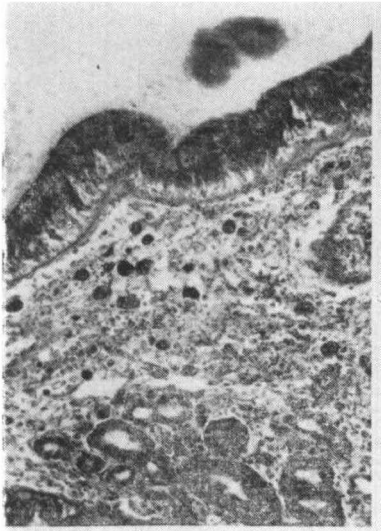
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Fig. 9. Section of stomach epithelium, connective tissue and a portion of the digestive gland. The large dark bodies are the "pigment" cells filled with coarse granules of (natural) yellow color. Numerous parasites are present with the usual concentrations of leucocytes and fibroblasts. The tubules of the digestive gland show the characteristic shrinkage of the epithelium. Heidenhain's haematoxylin and eosin. Magnification about 220 diameters.

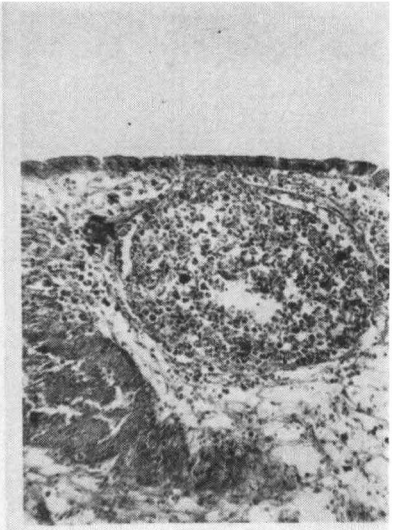
Fig. 10. Margin of an abscessed mantle. A characteristic abscess is against the epithelium and contains massed parasites, tissue debris, and leucocytes. Several smaller abscesses are shown also, which are filled with the peculiar basophilic-staining fluid also shown in Fig. 8. Parasites are not confined to the abscesses; they are distributed throughout the surrounding connective tissue but in smaller numbers. Delafield's haematoxylin and eosin. Original magnification about 220 diameters.

Fig. 11. Abscesses in the mantle. The epithelium over one of these has disintegrated and that portion remaining is necrotic. "Pigment" bodies and fibrous tissue are also shown. Delafield's haematoxylin and eosin. Original magnification about 220 diameters.

Fig. 12. Section of a portion of mantle showing almost continuous layer of necrotic abscesses under the epithelium, and in the center the mantle artery containing large concentrations of parasites in the walls and lumen. Against one end of the artery is a large nerve trunk. The usual fibrosed condition and "pigment" bodies are shown. Note the large number of parasites in the lumen of the artery. Delafield's haematoxylin and eosin. Original magnification about 100 diameters.



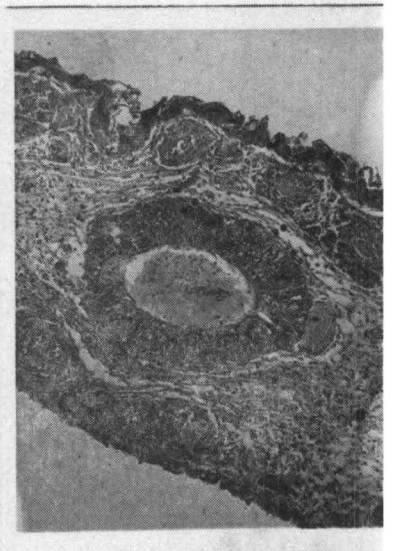
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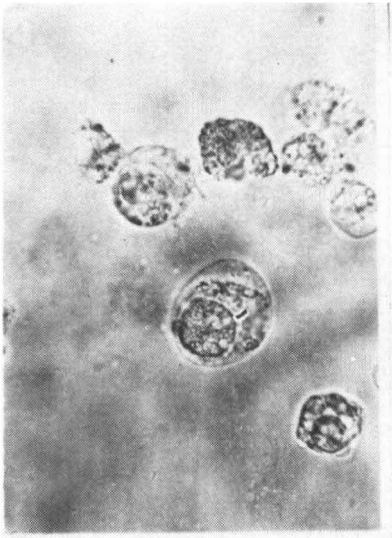
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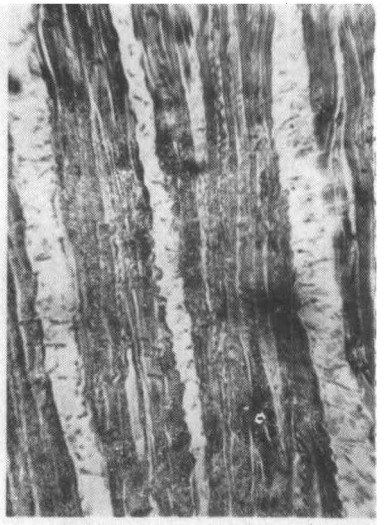
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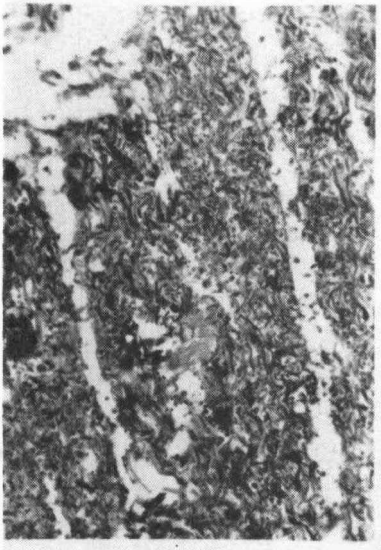
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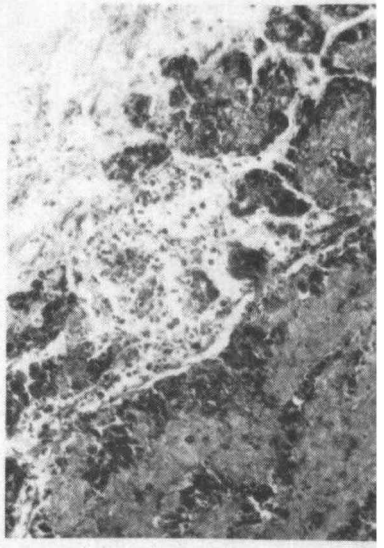
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epithelium which ends in necrosis of the cells. Compare Figure 6 which shows normal tubules, with Figure 9, a heavily infected tissue.

Peripheral nerves are seldom penetrated by *D. marinum* even when concentrations of parasites may be heavy around the bundles (see Fig. 12, nerve at one end of oval blood vessel). A special study of the visceral ganglion however, shows that this primary nerve center is often invaded, especially in the cortical layer containing the nerve cell bodies or neurons proper. Small abscesses occur in this layer with complete destruction of nerve cells (Fig. 16).

Considerable study of the effects of *D. marinum* infection on development of gonadal tissues has been made. Because of the great interest in this phase of the matter and the considerable data involved, this part of the investigation will be presented in a later paper.

DISCUSSION. While much remains to be done in analysis of effect of *D. marinum* infection of oysters, certain broad points of interest are apparent. The first of these is that the disease has several points of similarity of effect to certain generalized mycotic infections in man. The fact of susceptibility of nearly any tissue to invasion is distinctly like fungus infections of the type of histoplasmosis and North American blastomycosis. Widespread involvement of various organs and tissues without development of acute toxic symptoms, and steady progressive development, are characteristic of the human mycotic diseases in which the infective element is a spherical yeast-like cell not develop-

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Fig. 13. Moribund leucocytic cells from the pericardial cavity of a gaper. The tissue debris from disintegrating cells is also characteristic. *D. marinum* cells are contained in the leucocytes. Original magnification about 950 diameters.

Fig. 14. Longitudinal section of bundles of muscle fibers from the adductor muscle of a normal, uninfected oyster. The blood sinuses are traversed by a light fibrous tissue. Note length of normal fibers as shown by several in contraction. Delafield's haematoxylin and eosin. Magnification about 220 diameters.

Fig. 15. Lysis and destruction of a heavily infected portion of adductor muscle. Original location of two blood sinuses may still be seen. Note irregular masses of short sections of fibers disconnected from each other. Delafield's haematoxylin and eosin. Original magnification about 220 diameters.

Fig. 16. Section of the margin of the visceral ganglion of a heavily infected gaper. The darkest cells are neurons. An abscessed area containing numerous *D. marinum* cells has cut through the cortex and a few scattered parasites may be seen in small lysed cavities deeper in the medullary portion of the ganglion. Delafield's haematoxylin and eosin. Original magnification about 220 diameters.

ing mycelia in the host. In addition to those named above, South American blastomycosis, cryptococcosis, coccidioidomycosis and rhinosporidiosis belong in the same category. This is not a suggestion that mycotic infection of man may result from *D. marinum;* there are no data to indicate such relation.

The capacity of *D. marinum* to build up massive infections before death of the host is also of interest. This would seem to indicate that the association of the two species may be of long standing, allowing as it does to both host and parasite a period of reproduction before infection destroys the host. While the exact percentages of infection for oysters of less than one year have not been worked out as meticulously as have those for older oysters, it has been obvious from the beginning that such younger individuals are not infected to the extent that older market size individuals are. It is not believed that this is a physiological difference in susceptibility, but probably is a result of the mechanics of infection. However, oysters in Louisiana under one year of age are vigorously reproductive and as far as interrelationship of species is concerned there seems to be a successful adjustment between parasite and host. This adjustment has not been carried to the point of tolerance, by individual hosts, of heavy infection, indeed there seems to be no effective physiological brake, in the form of immune reaction, to the progressive lysis of entire organs and systems.

In conclusion, the tendency of *D. marinum* infection to produce multiple abscesses in varied tissues is of interest. A study of the literature of oyster mortalities shows that abscess production is one of the common denominators wherever sufficiently minute observations have been made. The studies of Giard (1894), Needler and Logie (1947), and Roughley (1926) are cases in point. It is not being suggested that *D. marinum* is the etiological agency in those cases, but that the waves of shellfish mortality in general may often be produced by organisms of similar physiological nature and that the preponderance of effort devoted by biologists to environmental measurement without correlated pathological studies may be misdirected.

SUMMARY

A study of the histopathological effects of infection of oysters by *Dermocystidium marinum* shows that the early stages are characterized by inflammation followed in turn by fibrosis and extensive lysis of the tissues. Under conditions of high temperature and high salinity

massive infections develop, involving all tissues but producing most extensive damage to connective tissues, adductor muscle, digestive epithelium and blood vessels. Attention is called to the resemblance of this type of infection to certain mycoses of man.

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